Computational Macro - Problem Set 2

Question 1.1: Optimality conditions

a) sequence problem: $v(a_t,z_t) = \sup_{\{a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t u(z_t + Ra_t - a_{t+1}) \mid z_t\right]$

↓

Bellman equation: $v(a_t,z_t) = \sup_{a_1}\left(\mathbb{E}_0[u(z_1 + Ra_0 - a_1) \mid z_1] + \sup_{\{a_{t+1}\}_{t=1}^{\infty}} \mathbb{E}_0\left[\sum_{t=1}^{\infty}\beta^t u(z_t + Ra_t - a_{t+1}) \mid z_t\right]\right)$

$$v^*(a,z) = \sup_{a'} u(z + Ra - a') + \beta\mathbb{E}[v^*(a',z') \mid z]$$

The state consists of the two predetermined variables (a,z):

- a determines output and hence consumption and savings opportunities today - and thus resources for tomorrow
- z determines a) the transition probabilities for the income tomorrow and b) income today

Together, a and z hence determine utility today and the expected consumption and savings options tomorrow, i.e. expected future utility. The discount and interest rate are constant over time, so we do not need to track t. a and z hence summarise all the relevant information needed to choose consumption (and savings) today.

b) FOC [a']: $\frac{\partial}{\partial a'} v(a,z) = -u_c(z + Ra - a') + \beta\mathbb{E}[v_a(a',z') \mid z] \stackrel{!}{=} 0$

c) Envelope condition:

$$v(a,z) = u(z + Ra - g(a,z)) + \beta\mathbb{E}[v(g(a,z),z') \mid z] \quad \Big| \frac{d}{da}$$

$$v_a(a,z) = u_c(z + Ra - g(a,z)) \cdot (R - g_a(a,z)) + \underbrace{\beta\mathbb{E}[v_a(g(a,z),z')\, g_a(a,z) \mid z]}$$

as $g_a(a,z)$ is deterministic and independent of z': $= \beta g_a(a,z)\mathbb{E}[v_a(g(a,z),z') \mid z]$

$$= g_a(a,z)\underbrace{\left(-u_c(z + Ra - g(a,z)) + \beta\mathbb{E}[v_a(g(a,z),z') \mid z\right)}_{=0 \text{ by FOC}} + Ru_c(z + Ra - g(a,z))$$

$$= Ru_c(z + Ra - g(a,z))$$

d) Euler equation

Plug Envelope condition into FOC[a']:

$$u_c(z + Ra - a') = \beta\mathbb{E}[Ru_c(z' + Ra' - a'') \mid z] \quad \Big| R \text{ is deterministic}$$

$$\Leftrightarrow u_c(c) = \beta R\mathbb{E}[u_c(c') \mid z]$$

Question 1.2: The analytical household of Acharya and Dogra

a) Euler equation: $\frac{\partial}{\partial c}\left(-\frac{1}{\gamma}e^{-\gamma c}\right) = e^{-\gamma c}$, so

$$e^{-\gamma c} = \beta R\mathbb{E}[e^{-\gamma c'} \mid z]$$

$$\Leftrightarrow e^{-\gamma c} = \beta R\mathbb{E}[e^{-\gamma c'} \mid x,e] \qquad z = \bar{z} + e, \quad x = z + Ra$$

BC: $c + a' = z + Ra \Rightarrow a' = \underbrace{z + Ra}_{x} - c = x - c$

Consumption tomorrow:

$$\begin{aligned}
c' &= \mu x' + v e' + f && | \; x' = z' + Ra' \\
&= \mu(z' + Ra') + v e' + f && | \; z' = \bar{z} + e' \;,\; a' = x - c \\
&= \mu(\bar{z} + e' + R(x-c)) + v e' + f \\
&= \mu(\bar{z} + e') + \mu R x - \mu R(\mu x + v e + f) + v e' + f \\
&= \mu\bar{z} + (\mu + v) e' + \mu R x (1-\mu) - \mu R v e + f(1 - \mu R) && | e' = \rho e + \sigma_\varepsilon \varepsilon' \\
&= \mu R(1-\mu) x + \mu(\bar{z} - Rf) + (\mu+v)(\rho e + \sigma_\varepsilon \varepsilon') - \mu R v e + f \\
&= \mu R(1-\mu) x + \mu(\bar{z} - Rf) + e((\mu+v)\rho - \mu R v) + (\mu+v)\sigma_\varepsilon \varepsilon' + f \\
&= \mu R(1-\mu) x + (\mu(\rho - Rv) + v\rho) e + \mu(\bar{z} - Rf) + f + (\mu+v)\sigma_\varepsilon \varepsilon' \;//
\end{aligned}$$

b) $\mathbb{E}(e^{-\gamma c'} | x, e) = \exp[\mathbb{E}(-\gamma c' | x, e) + \mathrm{Var}(-\gamma c' | x, e)/2]$

$= \exp[-\gamma \mathbb{E}(c' | x, e) + \gamma^2 \mathrm{Var}(c' | x, e)/2]$

$\mathbb{E}(c' | x, e) = \mu R(1-\mu) x + [\mu(\rho - Rv) + v\rho] e + \mu(\bar{z} - Rf) + f$

$\mathrm{Var}(c' | x, e) = (\mu+v)^2 \sigma_\varepsilon^2$

together: $\mathbb{E}(e^{-\gamma c'} | x, e) = \exp[-\gamma(\mu R(1-\mu) x + [\mu(\rho - Rv) + v\rho] e + \mu(\bar{z} - Rf) + f) + \gamma^2/2 \; (\mu+v)^2 \sigma_\varepsilon^2]$

Euler equation: $e^{-\gamma c} = \beta R \, \mathbb{E}(e^{-\gamma c'} | x, e)$ | log

$-\gamma c = -\gamma(\mu R(1-\mu) x + [\mu(\rho - Rv) + v\rho] e + \mu(\bar{z} - Rf) + f)) + \gamma^2/2 \; (\mu+v)^2 \sigma_\varepsilon^2 + \ln(\beta R)$ | $: -\gamma$

$$c = \mu R(1-\mu) x + [\mu(\rho - Rv) + v\rho] e + \mu(\bar{z} - Rf) + f - \frac{\gamma}{2}(\mu+v)^2 \sigma_\varepsilon^2 - \frac{\ln(\beta R)}{\gamma}$$

Matching coefficients: • $\mu = \mu R(1-\mu) \overset{\mu \neq 0}{\Leftrightarrow} 1 = R(1-\mu) \Leftrightarrow \frac{1}{R} = 1 - \mu \Leftrightarrow \mu = 1 - \frac{1}{R}$

$\Rightarrow \mu = \frac{R-1}{R} = \frac{r}{R}$ //

• $v = \mu(\rho - Rv) + v\rho \Leftrightarrow v(1 - \rho + \underbrace{\mu R}_{r}) = \mu\rho$ | $1 + r = R$

$\Rightarrow v = \frac{\rho\mu}{R - \rho}$ //

• $f = \mu(\bar{z} - Rf) + f - \frac{\gamma}{2}(\mu+v)^2 \sigma_\varepsilon^2 - \frac{\ln(\beta R)}{\gamma}$ | $1 - f$

$\mu R f = \mu\bar{z} - \frac{\gamma}{2}(\mu+v)^2 \sigma_\varepsilon^2 - \frac{\ln(\beta R)}{\gamma}$ | $: \mu R = r$

$f = \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma(\mu+v)^2 \sigma_\varepsilon^2}{2r}$ //

c) Mu is the annuity factor, i.e. its marginal propensity to consume cash on hand. Annuity means that the "asset" (i.e. cash on hand here) is never run down but only the interest is consumed, so additional consumption is perfectly smoother over time. Out of 1 additional unit of cash hand, the household consumes r/R and saves 1/R. Reversely, a flow of r/R every period from today to infinity has present value 1.
This means that MPC is low, and it is independent of risk or impatience (i.e. not on gamma, beta, sigma_epsilon).

d) $\frac{\partial e}{\partial \sigma_\varepsilon \varepsilon} = \frac{\partial}{\partial \rho}\left(\rho e_{-1} + \sigma_\varepsilon \varepsilon\right) = 1$

$\frac{\partial x}{\partial \sigma_\varepsilon \varepsilon} = \frac{\partial}{\partial \rho}\left(z + Ra\right) = \frac{\partial}{\partial \rho}\left(\bar{z} + c + Ra\right) = \frac{\partial}{\partial \rho}\left(\bar{z} + \rho e_{-1} + \sigma_\varepsilon \varepsilon + Ra\right) = 1$

Together: $\frac{\partial c}{\partial \sigma_\varepsilon \varepsilon} = \frac{\partial}{\partial \sigma_\varepsilon \varepsilon}(\mu x) + \frac{\partial}{\partial \sigma_\varepsilon \varepsilon}(\nu e) + \frac{\partial}{\partial \sigma_\varepsilon \varepsilon} f = \mu \underbrace{\frac{\partial x}{\partial \sigma_\varepsilon \varepsilon}}_{1} + \nu \underbrace{\frac{\partial e}{\partial \sigma_\varepsilon \varepsilon}}_{1} = \mu + \nu$

($\frac{\partial \mu}{\partial \sigma_\varepsilon \varepsilon} = 0$, $\frac{\partial \nu}{\partial \sigma_\varepsilon \varepsilon} = 0$, f constant)

$\mu + \nu = \frac{r}{R} + \frac{\rho \mu}{R - \rho} = \frac{r}{R} + \frac{r}{R}\frac{\rho}{R-\rho} = \frac{r}{R}\left(1 + \frac{\rho}{R-\rho}\right) = \frac{r}{R}\left(\frac{R-\rho+\rho}{R-\rho}\right)$

$= \frac{r}{R-\rho}$ //

Effect of persistence: $\frac{\partial}{\partial \rho}(\mu + \nu) = \frac{r}{(R-\rho)^2} > 0$ (assuming $r > 0$)

$\frac{\partial}{\partial \rho}\frac{r}{(R-\rho)^2} = \frac{2r}{(R-\rho)^3} > 0$ (as $R > \rho$ by $\rho \in [0,1)$)

$\Rightarrow$ convex

$\lim_{\rho \to 0} \frac{\partial}{\partial \rho}(\mu + \nu) = \frac{r}{R}$

$\lim_{\rho \to 1} \frac{\partial}{\partial \rho}(\mu + \nu) = \frac{r}{1 + r - 1} = 1$

The marginal propensity to consume out of an income surprise responds positively to persistence in the income process, i.e. if an income process is more persistent (i.e. is more similar to last period's income) the household consumes more out of an income process.
If persistence goes to 0, the response goes to r/R (=mu), i.e. if income is basically a random walk, the household consumes the annuity value of the income shock and the windfall is smoothed for the infinite future.
If persistence goes to 1, the response goes to 1, i.e. if income is most certainly the same as last period, consumption goes up by the same amount this (and every future) period, as in expectation, the household will have this extra amount of resources available every period from now on.

e) $\rho = 0 \Rightarrow \mu = \frac{r}{R}$, $\nu = 0$, $f = \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma \mu^2 \sigma_\varepsilon^2}{2r}$

$c|_{\rho=0} = \frac{r}{R} x + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{\gamma r \sigma_\varepsilon^2}{2R^2}$

With an i.i.d. income process, the consumption policy does not include the shock e, it only depends on cash on hand*. The shock does not say anything about predicted income, so it is not relevant for the consumption choice anymore.

* x does include e, but e is not relevant for the consumption policy anymore